

WEIYI PAN

Postdoctoral researcher, University of Regensburg, Germany

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ACADEMIC CAREER

B.Sc. in Physics, Huazhong University of Science and Technology, China 2016.09 - 2020.06

Ph.D. in Physics, Tsinghua University, China 2020.09-2025.06

- Thesis: Theoretical study of two-dimensional magnets: From magnetic interactions to novel spin textures (Outstanding Doctoral Thesis in Tsinghua University for 2025)
- **Supervisor: Prof. Dr. Wenhui Duan**
(Fellow of the American Physical Society (APS), Academician of the Chinese Academy of Sciences (CAS))
- Collaborate with **Dr. Changsong Xu** at Fudan University
- GPA rank: 3/75

Postdoctoral researcher, University of Regensburg, Germany 2025.08-now

- Research theme: Theoretical spintronics
- **Supervisor: Prof. Dr. Jaroslav Fabian**
(Spokesperson of Collaborative Research Center CRC 1277 ‘Emergent relativistic effects in condensed matter’ at Uni. Regensburg)

PUBLICATIONS

- **Wei Yi Pan**, Changsong Xu, Xueyang Li, Zhiming Xu, Boyu Liu, Bing-Lin Gu and Wenhui Duan, Chiral magnetism in lithium-decorated monolayer CrTe₂: Interplay between Dzyaloshinskii-Moriya interaction and higher-order interactions. *Phys. Rev. B* 109, 214415 (2024)
- **Wei Yi Pan**, Shilei Ji, and Zhiming Xu, Strain-Induced Intrinsic Antiferromagnetic Skyrmions in Two-Dimensional Janus Magnets, *Phys. Rev. B* 110, 144408 (2024)
- **Wei Yi Pan**, Xueyang Li, and Junsheng Feng, Strain-induced frustrated magnetism and topological spin configurations in LiCrTe₂, *Phys. Rev. B* 108, 224417 (2023)
- **Wei Yi Pan**, Tuning the magnetic anisotropy and topological phase with electronic correlation in single-layer *H*-FeBr₂, *Phys. Rev. B* 106, 125122 (2022)
- Shilei Ji, **Wei Yi Pan**, Chuye Quan, Xiaoyang He, Chen Chen, Jianping Yang, and Xing’ao Li, Valley-contrasting transport of the coupling between spin and orbital angular momentum in ferromagnetic transition metal dichalcogenides, *Phys. Rev. B* 110, 075418 (2024)
- **Wei Yi Pan**, et al., Long-Range Spin-Orbit-Coupled Magnetoelectricity in Type-II Multiferroic NiI₂, arXiv: 2502. 16442(2025)
- Haitao Wang*, Tianxing, Jiang*, **Wei Yi Pan***, et al., Microscopic evidence of spin-driven multiferroicity and topological spin textures in monolayer NiI₂, ***Equal contribution. Under review at Phys. Rev. Lett..**

PRESENTATIONS AND TALKS

- Oral presentation, the conference of PFUNT (A physics conference held by the top 5 university in China) 2024, Hefei, China, Dec 2024
(**2nd-prize award for oral presentation**)
- Poster presentation, the conference of condensed matter physics (CCMP) 2024, Liyang, China, Aug 2024
- Poster presentation, International Conference on Magnetism 2024, Bologna, **Italy**, Jul 2024
- Oral presentation, the conference of PFUNT 2023, Shanghai, China, Dec 2023
(**3rd-prize award for oral presentation**)
- Poster presentation, Chinese Physical Society (CPS) fall meeting 2023, Yinchuan, China, Aug 2023
(**Best poster presentation awarded**)

HONORS AND AWARDS

- **Yu-Hsun Woo Scholarship**, Department of Physics, Tsinghua University, Jun 2025
(Highest Honor for Graduating PhD candidates with best academic performance)
- **Outstanding Doctoral Thesis Award for Tsinghua University**, Jun 2025
(Awarded to **top 1%** of graduates for the whole university)
- **First Prize for Oral Presentation, PhD Academic Symposium**, Department of Physics, Tsinghua University, 2022, 2023
(Awarded twice for exceptional presentation skills at annual doctoral forums)
- **Comprehensive First-Class Scholarship**, Tsinghua University, 2022, 2023, 2024
(Awarded three times for **top 1%** academic performance within each year)

RESEARCH INTERESTS

My research lies in the theoretical understanding of magnetism and spintronics-related phenomena in realistic systems, especially those emerging two-dimensional magnets. Currently, I have involved in the following points:

- Spin-related response, transport properties, and spin-orbit torques in realistic systems.
- First-principles exploration of novel magnetic interactions: Higher-order interactions, Kitaev interactions, chiral interactions.
- Microscopic origin of exotic noncollinear spin textures: Skyrmions, spin spiral, frustrated magnetism.
- Spin-induced ferroelectric and multiferroic properties, magnetoelectric coupling.

RESEARCH SKILLS

Skill 1: Electronic structure theory

- Involving first-principles calculation and Wannier-function-based tight-binding modeling for realistic systems.

Skill 2: Quantum transport theory

- Involving evaluating the response coefficient based on Kubo formula and semiclassical transport theory.

Skill 3: Magnetic interactions and spin models

- Involving calculating the magnetic couplings and thus construct the spin model for realistic magnets.

Skill 4: Atomistic spin simulations

- Simulating the classical spin configurations in realistic magnets based on the spin model.

Programming skills: Fortran, Python, MATLAB